Sales Performance & Revenue Analysis Across Customers, Regions and Sales Representatives

1. Project Overview and Objective

This project aims to analyze sales transaction data to uncover insights related to revenue generation, customer purchasing behavior, and sales representative performance. The analysis focuses on identifying key revenue drivers, evaluating sales trends, and supporting data-driven decision making.

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2. Data Sources

- **Source Description and Timeline:** Sales transaction dataset containing detailed information about products, customers, sales representatives, payment methods, and revenue **and** 2013.
- **Domain**: Retail / Sales Analytics

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3. Problem Statement

• Analyze overall sales revenue and transaction volume

• Identify top-performing sales representatives

• Evaluate revenue contribution by region

• Analyze customer purchasing behavior (New vs Returning customers)

• Examine product category performance

• Evaluate discount impact on revenue

• Analyze sales trends over time

• Forecast future sales trends

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4. Attribute (Column /Features) Details:

Raw Data:

Attribute Name	Data Type	Description
Product_ID	Whole Number	Unique ID for product
Sale_Date	Date	Date in which sales happened
Sales_Rep	Text	Name of the sale rep
Region	Text	Region in which sales happened
Sales_Amount	Decimal Number	Total sales amount
Quantity_Sold	Whole Number	quantity of the product sold
Product_Category	Text	category of the product
Unit_Cost	Decimal Number	cost per product
Unit_Price	Decimal Number	price per product
Customer_Type	Text	type of customer
Discount	Decimal Number	discount applied on product
Payment_Method	Text	mode of payment
Sales_Channel	Text	sales platform
Region_and_Sales_Rep	Text	concatenation of sales rep and region

Sales_data (Fact Table)

Attribute Name	Data type	Description
Product_ID	Whole Number	Unique ID of product
Sale_Date	Date	Date in which Sales happened
Rep_ID	Text	Unique Id for sales rep
Region	Text	Region in which sales happened
Quantity_Sold	Whole Number	number of units sold
Category_ID	Text	Unique id of category
Unit_Price	Decimal Number	Price per product
Customer_Type	Text	type of customer
Discount	Decimal Number	discount applied on product
Payment_Method	Text	Mode of payment
Sales_Channel	Text	sales platform

Product_Category(dimension table)

Attribute Name	Data Type	Description
Category_ID	Text	Unique id of category
Product_Category	Text	Name of the category

Sales_Rep(dimension table)

Attribute Name	Data Type	Description
Rep_ID	Text	Unique Id for sales rep
Rep_Name	text	Name of Sale rep

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5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

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6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created calculated fields.
  - Removed few columns which are not useful for analysis
  - Created dimension tables (Product_Category, Sales_Rep)
  - Standardized data format for all the columns

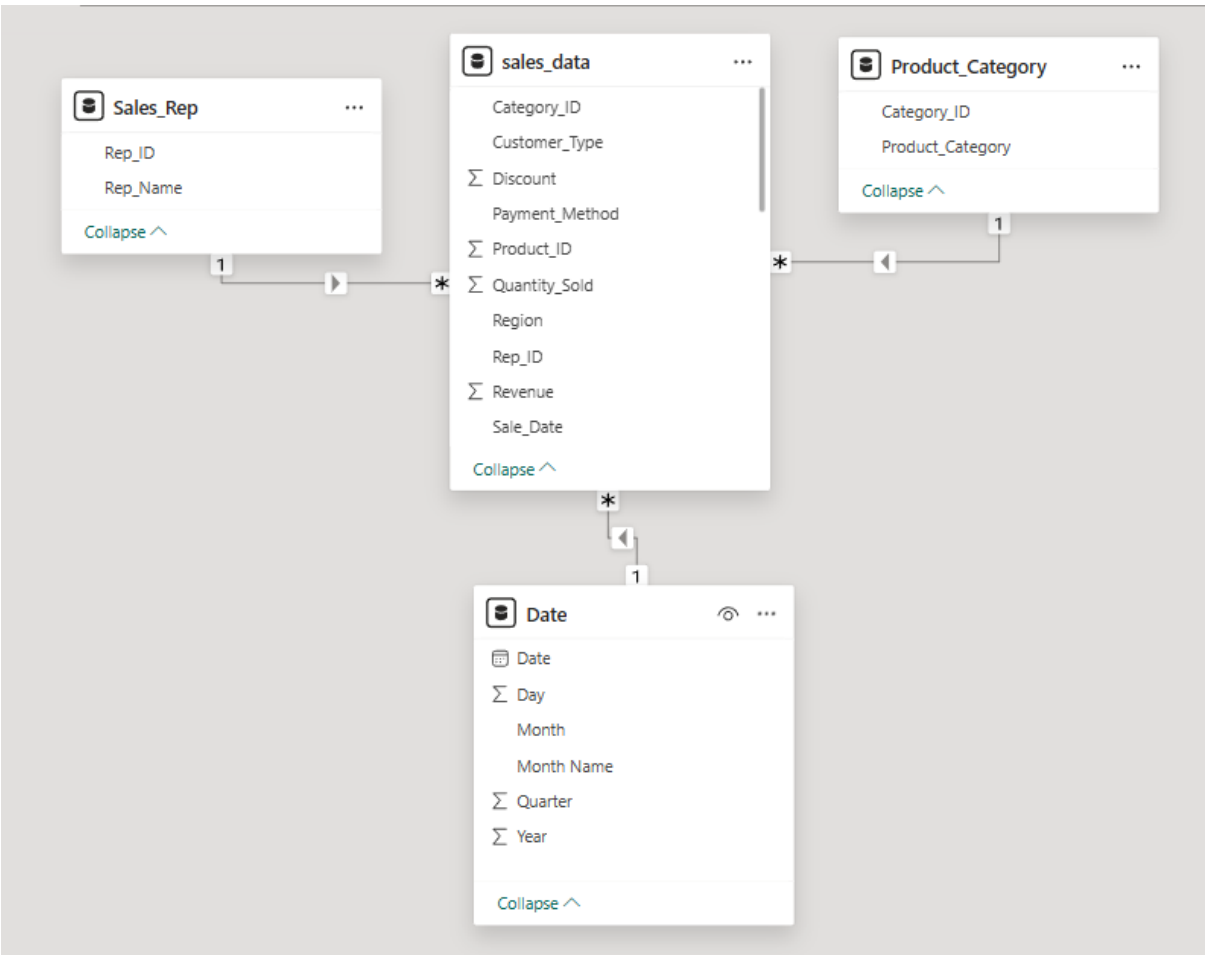
7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.
- **Created a data dimension table**

- **Created calculated column**

  Revenue = Unit_price*Quantity_Sold*(1-Discount)

sales_data (Category_ID)	* — 1	Product_Category (Category_ID)	Active
sales_data (Rep_ID)	* — 1	Sales_Rep (Rep_ID)	Active
sales_data (Sale_Date)	* — 1	Date (Date)	Active

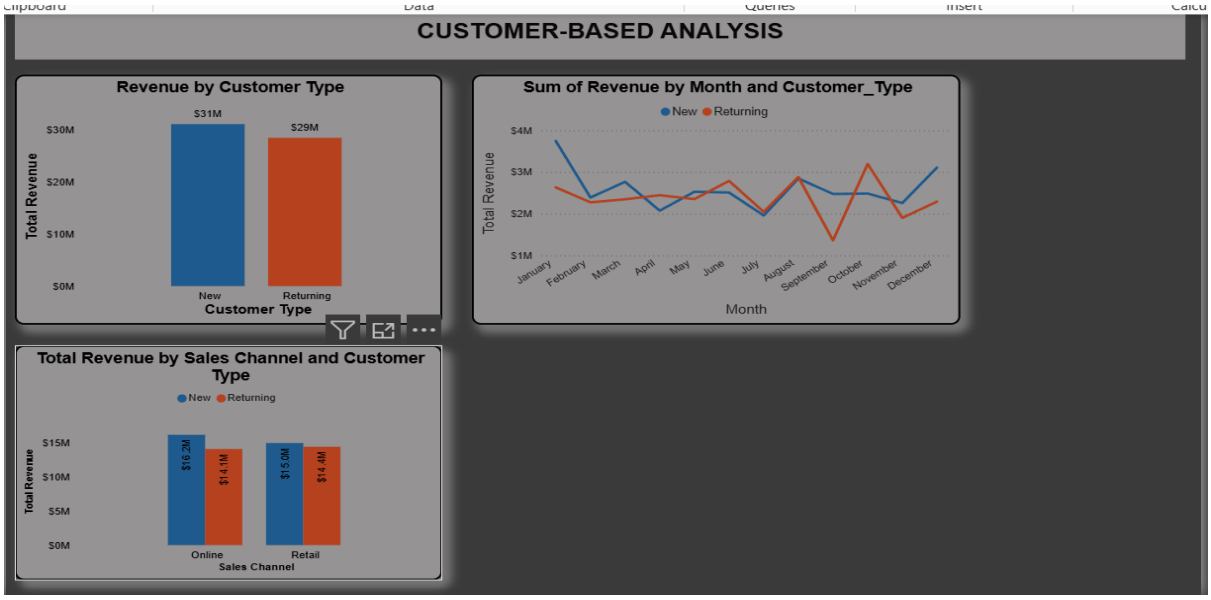


- **Calculated Columns & DAX Measures:** Implemented DAX formulas for key metrics, such as total sales, growth percentages, and cumulative totals**. (List Out)**
  1. Average_Discount = AVERAGE(sales_data[Discount])
  2. previous_month =CALCULATE(SUM(sales_data[Revenue]),PREVIOUSMONTH('Date'[Date]))
  3. Total_Quantity = SUM(sales_data[Quantity_Sold])
  4. Total_Sales = SUM(sales_data[Revenue])
  5. MOM% = VAR PrevMonth = CALCULATE([Total_Sales],DATEADD('Date'[Date], -1, MONTH)) RETURN DIVIDE([Total_Sales] - PrevMonth, PrevMonth)

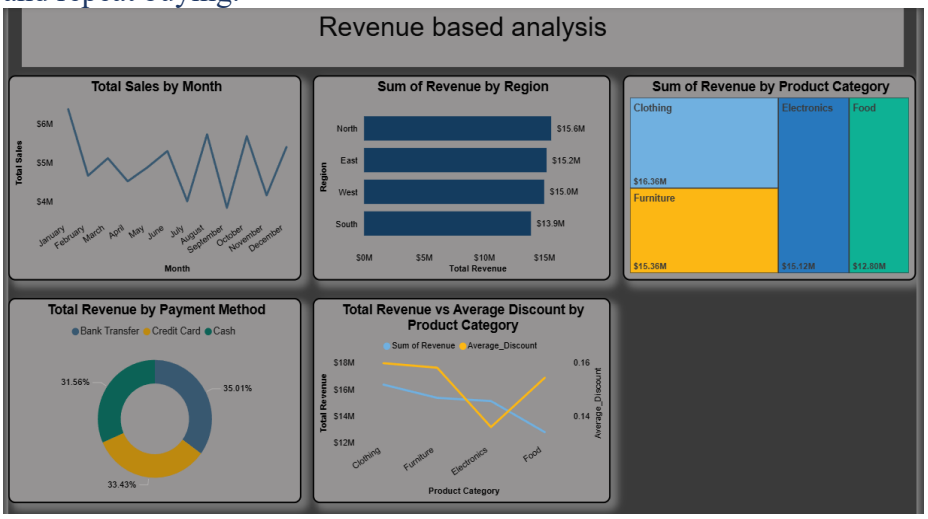
8. Analysis and Visualizations (Power BI)

Dashboard Features:

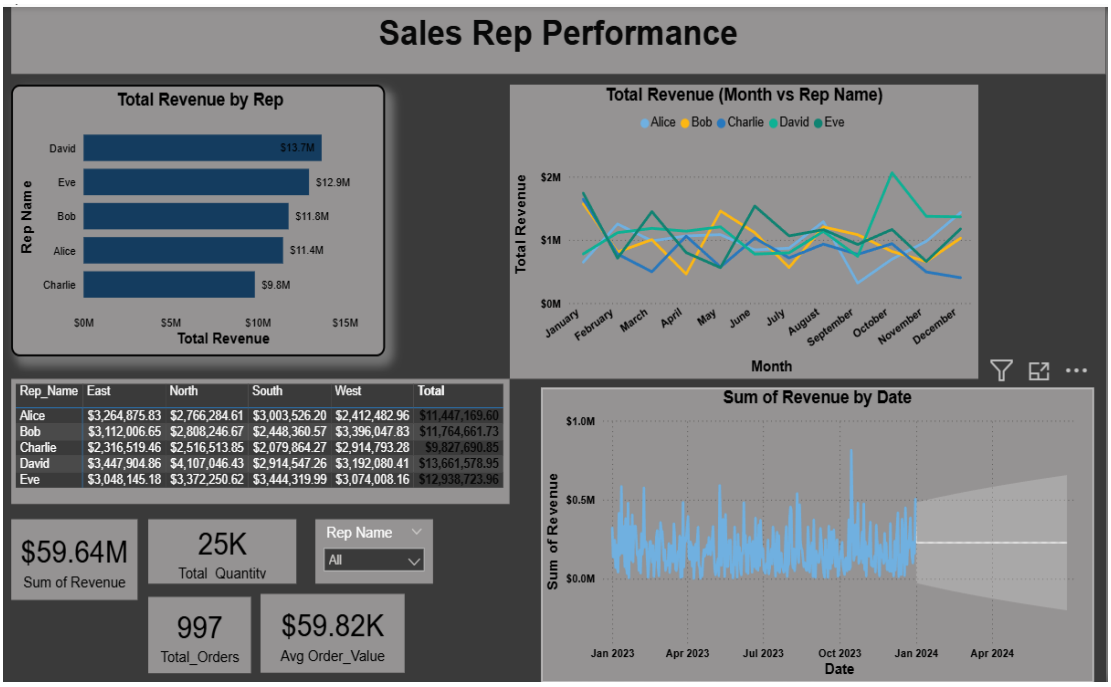
- **Multiple Visualizations based on problem statement:** Bar charts, line charts, pie charts, cards, and tables to communicate insights. **(List Out)**



- The business is successfully attracting new customers while maintaining a strong returning customer base.
- New customer revenue peaks in January and December, suggesting strong seasonal demand at the beginning and end of the year.
- Returning customers show fluctuations, with a sharp drop around September followed by recovery in October.
- Online sales are the primary revenue driver, especially for new customer acquisition.
- New customers contribute more revenue than returning customers across both sales channels.
- Returning customers show consistent purchasing patterns, indicating loyalty and repeat buying.



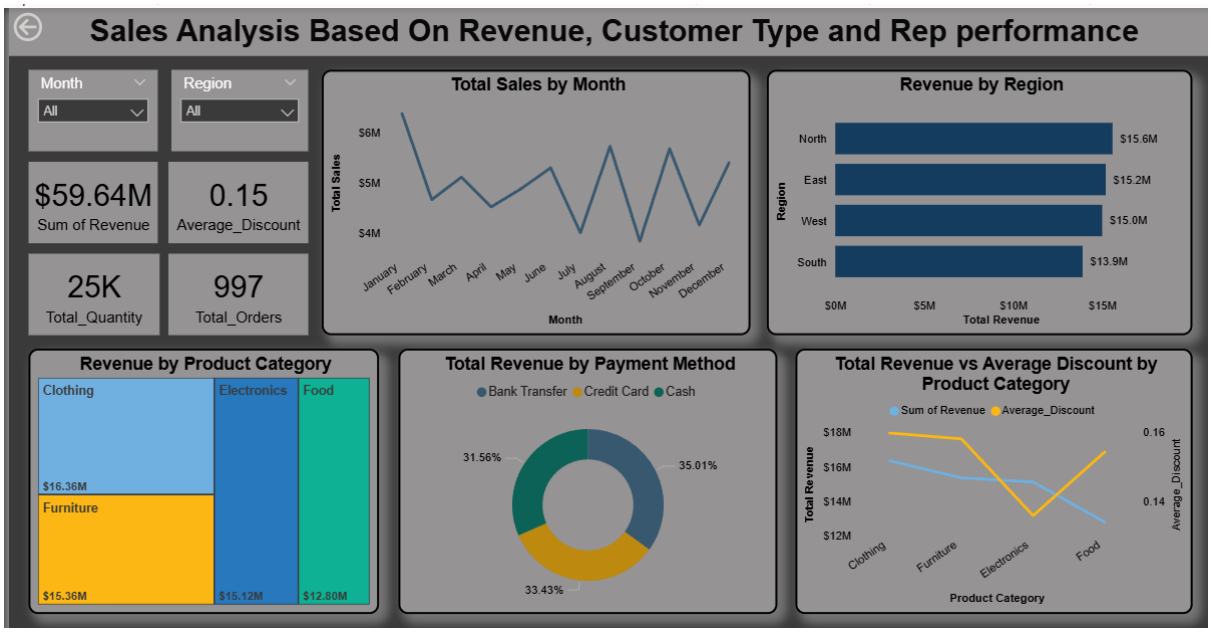
- Revenue shows seasonal fluctuations throughout the year, with peaks in January and August, while July and October show noticeable drops, indicating possible seasonal demand patterns.
- North region generates the highest revenue ($15.6M) followed closely by East and West, while the South region contributes the least ($13.9M), suggesting an opportunity to improve sales performance in the South.
- Clothing is the top-performing product category ($16.36M), followed by Furniture and Electronics, whereas Food generates the lowest revenue ($12.8M) among all categories.
- Payment methods are evenly distributed, with Bank Transfer (35.01%) slightly leading, followed by Credit Card (33.43%) and Cash (31.56%), indicating balanced customer payment preferences.
- Clothing generates the highest revenue (~$16M) even with a moderate discount level, indicating strong customer demand for this category.
- Electronics has the lowest average discount (~14%) but still maintains moderate revenue (~$15M), suggesting sales are less dependent on heavy discounting.
- Food category shows relatively higher discounts but generates the lowest revenue (~$13M), indicating discounts alone are not significantly boosting sales.
- Overall, there is no strong positive relationship between higher discounts and higher revenue, meaning product demand likely influences revenue more than discount levels.

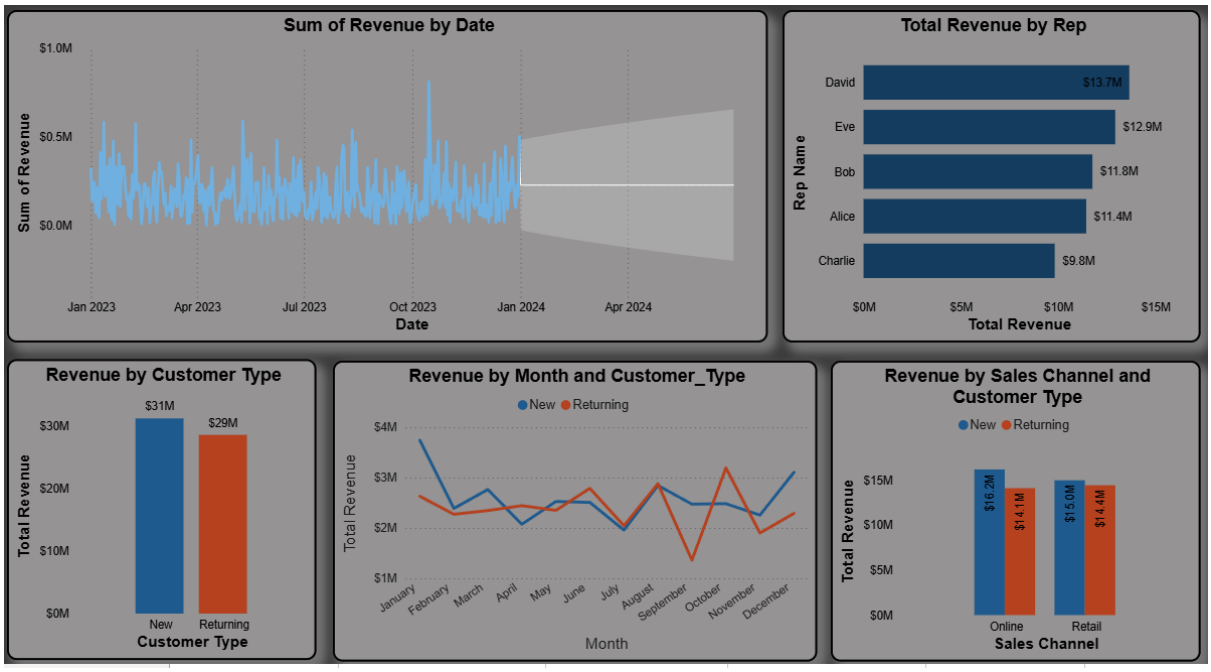


- David is the top-performing sales representative, generating the highest revenue of $13.7M, followed by Eve ($12.9M) and Bob ($11.8M), indicating strong individual performance differences among the reps.
- Charlie records the lowest revenue ($9.8M) compared to other sales representatives, suggesting potential opportunities for performance improvement or additional support.
- Sales performance varies month-to-month across representatives, with noticeable peaks for different reps during certain months, indicating seasonal demand and varying sales effectiveness.
- The North and East regions contribute significantly to overall revenue for most sales representatives, making them the strongest performing regions in terms of sales contribution.

- Overall business performance is strong, with $59.64M total revenue generated from 997 orders and 25K total quantity sold, resulting in an average order value of about $59.82K.

SCREENSHOT OF DASHBOARD /REPORT





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9. Insights & Conclusions

Key Findings:

- Total revenue generated is $59.64M from 997 orders and 25K units sold.
- North region generated the highest revenue, while South region generated the lowest.
- Clothing is the top-performing product category, while Food contributes the least revenue.
- Online sales channel generates slightly higher revenue compared to retail.

- New customers ($31M) contribute slightly more revenue than returning customers ($29M).
- David is the top-performing sales representative, while Charlie has the lowest revenue contribution.
- Bank Transfer, Credit Card, and Cash are almost equally used payment methods.
- Discounts show minimal impact on overall revenue generation.

Insights:

- Sales performance is fairly balanced across regions and product categories, reducing dependency on a single segment.
- The online channel plays a key role in driving sales growth.
- The business is effective at acquiring new customers, but improving retention could further increase revenue.
- Food category and South region present potential opportunities for growth.
- Differences in sales rep performance indicate opportunities for training or performance optimization.

10. Conclusions

The sales analysis shows that the business is performing well with $59.64M in total revenue, supported by strong contributions from multiple regions, product categories, and sales channels. The North region and Clothing category are the major revenue drivers, while the online channel slightly outperforms retail, indicating the importance of digital sales.
The company is successful in attracting new customers, but improving customer retention could further enhance revenue growth. Additionally, discounts do not significantly impact sales, suggesting that product demand and sales strategies play a larger role in driving revenue.
Overall, the business has a balanced revenue distribution, with opportunities to improve performance in the South region, Food category, and lower-performing sales representatives to achieve further growth.